Alfien Jessurun P

ep23btech11001@iith.ac.in | alfienjessurun@gmail.com | +91 9566601594 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.56
St.Thomas Mat.Hr.Sec.School,Thoothukudi	Class XII (Tamilnadu SSLC)	2023	93.80%
St.Thomas Mat.Hr.Sec.School,Thoothukudi	Class X (Tamilnadu SSLC)	2021	100.00%

POSITIONS OF RESPONSIBILITY

Served as **Core Game Developer at Glitch (2025 -2026)**, engineering interactive sports mini-games for Milan'25 to drive campus-wide player engagement and retention.

Represented the institute as a core member of the official **chess** contingent across **two tenures** (2023-24 & 2025-26) at the **Inter-IIT Sports Meet**.

SKILLS

Programming Languages: C, C++, C#, Python, JavaScript, HTML5, Dart

Backend Frameworks: Node.js, Express.js, .NET Core, FastAPI

Databases & Storage: PostgreSQL, Relational Data Management

Real-Time & Caching: Socket.IO, Redis, In-Memory Caching, Low-Latency Architecture

APIs & Protocols: RESTful API Design, GraphQL, HTTP/2, JWT Authentication

DevOps & Testing: Jest, Supertest, AWS EC2, GitHub Actions CI/CD, Cloudflare containers.

App developement: Flutter, Android

GenAI & Agents: RAG Pipeline Architecture, Agentic AI Workflows, LLM Orchestration

Deep Learning & CV: Transformer Models, Computer vision

Data & Analytics: Supervised/Unsupervised Machine Learning, Statistical Data Analytics, Feature Engineering

Problem Solving: Solved 200+ problems on **Leetcode** and 75+ Problems in **Codeforces**

RELEVANT COURSES

Computer science courses: Introduction to Programming, Data Structures and Applications.

Machine learning courses: Data Science and Applications, Introduction to Machine learning.

PROJECTS

SubPlug - Subscription Intelligence Engine

Tech Stack: .NET 10, Python, LangGraph, PyTorch, PostgreSQL (pgvector), Docker, Cloudflare Containers, GitHub Actions

- Architected a 10-node **LangGraph** scoring graph and 7-stage resolution engine using **BGE embeddings** and

pgvector.
- Built an autonomous agent combining **web search** and **LLMs** to dynamically generate subscription cancellation guides.
- Containerized a dual **.NET/Python runtime** on **Cloudflare Containers** with automated **GitHub Actions CI/CD pipelines.**

Multi-Degradation Image Restoration Model (Hackathon Entry)

Tech Stack: Python, PyTorch, Deep Learning, SCUNet, NAFNet

- Built a deep learning pipeline for **multi-degradation denoising** and **2× super-resolution**, achieving **27.28 PSNR** and **0.7826 SSIM**.
- Built a configurable loss pipeline to rapidly ablate **SCUNet & NAFNet** architectures across multiple loss functions and optimize performance.
- Applied **test-time augmentation** during inference, **improving PSNR by +0.13 dB** and output consistency across evaluation runs.

Searchless Chess Action-Value Transformer

Tech Stack: Python, PyTorch, CUDA, WandB, NumPy

- Architected a **15M-parameter searchless Transformer** with custom **FEN/UCI tokenization**, using **HL-Gaussian loss** to evaluate board states without lookahead.
- Achieved **50.1% Lichess puzzle accuracy** within local GPU constraints, using only 1GB of data compared to DeepMind's 20GB+ dataset.
- Optimized GPU training with **bf16 mixed precision, gradient accumulation, and chunked loading** to operate within a 12GB VRAM budget.

Fourier & Physics-Informed Neural Operators (FNO / PINO)

Tech Stack: Python, PyTorch, NumPy, Jupyter, SciPy

- Implemented FNO architectures using **Fourier transform convolutions** to map function spaces and solve **complex PDEs**.
- Engineered PINO models with **physics-informed PDE loss** constraints to enforce physical consistency.
- Built evaluation pipelines benchmarking **multi-step rollout accuracy**, and other FNO vs. PINO metrics.

INTERNSHIPS

Software Engineering Intern | 360 Technology

- Designed the complete backend MVP using **.NET 10** and **PostgreSQL**, building secure transaction ingestion pipelines via India's **Account Aggregator** framework.
- Built an auditable **subscription detection engine** combining deterministic scoring rules with a 10-node **LangGraph LLM** reasoning pipeline.
- Containerized a dual-runtime (.NET/Python) application deployed to **Cloudflare Containers** via automated **GitHub Actions CI/CD pipelines**.